INCH-POUND
MIL-V-24740(SH)
31 May 1990

MILITARY SPECIFICATION

VALVE, SAFETY: BOILER, REMOTE PILOT OPERATED, FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high-performance boiler safety valve with a remote, hydraulic, non-steam flow pilot assembly, for steam drum and superheater service at working pressures of 600 and 1500 pounds per square inch (lb/in^2).

1.2 Classification. Valves shall be provided in the classes and styles specified in 1.2.1 and 1.2.2, as specified in the contract or order (see 6.2).

1.2.1 Classes. Valves shall be of the following classes:

a. Class 1 – 600 lb/in^2 service
b. Class 2 – 1500 lb/in^2 service

1.2.2 Styles. Valves shall be of the styles specified in table I (see 3.3.1.2 and table II).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-V-3	Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of
MIL-S-196	Support Items, Accessories, and Kits, Mechanical; Packaging of
MIL-S-901	Shock Tests, HI (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-S-1222	Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws, and Nuts
MIL-P-15024	Plates, Tags, and Bands for Identification of Equipment
MIL-P-15024/5	Plates, Identification
MIL-R-17131	Rods and Powders, Welding, Surfacing
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services
FED-STD-H28/1	Section 1; Screw-Thread Standards for Federal Services, Nomenclature, Definitions, and Letter Symbols for Screw Threads
FED-STD-H28/2	Section 2; Screw-Thread Standards for Federal Services, Unified Inch Screw Threads – UN and UNR Thread Forms
FED-STD-H28/4	Section 4; Screw-Thread Standards for Federal Services, Controlled Radius Root Screw Threads, UNJ Symbol

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FED-STD-H28/6	Section 6; Screw-Thread Standards for Federal Services, Gages and Gaging for Unified Screw Threads – UN and UNR Thread Forms
FED-STD-H28/19	Section 19; Screw-Thread Standards for Federal Services, Miscellaneous Threads
FED-STD-H28/21	Section 21; Screw-Thread Standards for Federal Services, Metric Screw-Threads

MILITARY

MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification
MIL-STD-278	Welding and Casting Standard
MIL-STD-777	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 **Other Government documents.** The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

NAVSHIPS 5000-S5103-841622	Valves, Boiler Safety, Dimensional Standard for 300, 600, and 1500 Psi Nominal Pressure
NAVSHIPS 250-660-30	A Guide for Design of Shock Resistant Naval Equipment

(Application for copies should be addressed to the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 **Non-Government publications.** The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5	Pipe Flanges and Flanged Fittings; (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code	Section I.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47 Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 269	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service; (DOD adopted)
A 351	Standard Specification for Steel Castings, Austenitic, for High-Temperature Service; (DOD adopted)
A 479	Standard Specification for Stainless and Heat-Resisting Steel Bars, and Shapes for Use in Boilers and Other Pressure Vessels; (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5), in accordance with 4.3.

3.2 **Material.**

3.2.1 **General.** Materials incorporated into these valve assemblies shall be as specified herein. Materials used in the design and construction of components shall be as shown in table I.

TABLE I. *Materials.*

Part	Material	Applicable documents
Body	Corrosion-resisting steel	ASTM A 351, CF8 or CF8C, ASME Boiler and Pressure Vessel Code
Bonnets	Corrosion-resisting steel	ASTM A 351, CF8 or ASTM A 479, grade 304, ASME Boiler and Pressure Vessel Code
Nozzles and disks, except superheater applications	Hard surfaced	MIL-R-17131, type MIL-RCoCr-A and hard surfacing requirements of MIL-STD-248
Nozzles and disks, super-heater	Stellite seating surfaces	MIL-STD-248
Threaded fasteners	Chromium-molybdenum vanadium steel	Bolts to B16 or better, nuts to grade 4 or better of MIL-S-1222
Unloader valve flanges	Steel	ANSI B16.5
Pilot valve	Corrosion-resisting steel	ASTM A 351, CF8 or ASTM A 479, grade 304, ASME Boiler and Pressure Vessel Code
Tubing and tubing couplings	Hard stainless steel	ASTM A 269

3.2.1.1 **Springs.** Springs shall be of non-corrosive materials that shall retain their properties under the service conditions specified herein (see 3.4.2.4 and 3.4.2.5).

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3.2.1.2 **Unloader valve.** The unloader valve parts shall be made of the material specified in table I. Equivalent or better materials proposed by the contractor will be considered for acceptance by the acquiring activity.

3.2.1.3 **Pilot assembly.** The pilot assembly may be subjected to continuous blowdown and therefore shall be constructed and its materials selected for withstanding full boiler water pressure and temperature in accordance with MIL-STD-777 (see ASTM A 351, CF8, or ASTM A 479, grade 304, ASME Boiler and Pressure Vessel Code).

3.2.2 **Recovered materials.** Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.3 **Material identification and control.** System fluid pressure retaining parts and the fasteners that provide the clamping force to retain these parts shall be subjected to strict material identification and control. The contractor shall assure the identification of chemical and physical properties of these parts by maintaining continuous traceability thereof to their original heat or batch. These parts shall be marked to relate the part to the material chemical composition and physical properties documentation. Nonmetallic materials and such items as setting or adjusting (threaded) parts 1/2-inch diameter or less shall be excluded from this requirement (see 6.3).

3.3 **Construction.** Construction of the safety valves shall be in accordance with 3.3.1 through 3.3.5.4.

3.3.1 **Characteristics.** Characteristics shall be in accordance with 3.3.1.1 through 3.3.1.4.

3.3.1.1 **Configuration.** The boiler safety valve system covered by this specification shall include the following general features, as defined hereinafter:

a. A safety valve (the unloader valve) for installation at the superheater outlet and the steam drum

b. A hydraulic, non-steam flow pilot assembly for continuously sensing boiler drum pressure; for actuating, automatically, the lifting of the unloader valve for blowdown; and for repressurization for safety valve closure

c. A boiler water storage canister, with filter, in the water supply line to the pilot valve assembly

d. Means for manually actuating lifting (popping) and reseating of the safety valve

e. The above items, other than the unloader valve, shall be located away from the vicinity of the boiler to permit ease of maintenance.

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Other appurtenances, to include a manually operated three-way valve and the self-contained piping between pilot assembly components, as necessary for satisfactory performance, shall be provided. Terminal piping and tubing connections for the sensing line, the water inlet line to the canister, and the actuating line shall be socket-welded ends located sufficiently away from their units in a convenient location for shipbuilder attachment of piping, as necessary. These lines shall contain suitable mechanical (takedown) fittings near the socket-welded ends.

3.3.1.2 **Dimensions.** NAVSHIPS 5000-S5103-841622 is referenced primarily to show general configuration of the unloader valve. The size of the safety (unloader) valve is the size of the inlet flange shown in table II (which is the nominal size of the inside diameter of the inlet opening at the flange face).

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TABLE II. *Dimensional and design data.*

Style	Inlet flange type (ANSI B16.5)	Inlet flange size (inch)	Outlet flange type (ANSI B16.5)	Outlet flange size (inch)	Drain flange type (ANSI B16.5)	Drain flange size (inch)	Drain flange location w/regard to outlet (degrees)	Flow rate (lb/hr)	Pop pressure (lb/in^2)	Steam temperature (°F)	Center to inlet (inch, plus or minus 1/16)	Center to outlet (inch, plus or minus 1/16)	Center to drain (inch, plus or minus 1/8)	Inlet to drain (inch, plus or minus 1/8)	Ship hull number
1	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 Counter-clockwise (CCW)	74,000	1350	950	7-3/4	7	6-11/16	6-3/32	CG-19-24
1a	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	72,000	1375	950	7-3/4	7	6-11/16	6-3/32	CG-29-31, 33
1b	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	74,000	1350	950	7-3/4	7	6-11/16	6-3/32	DDG-37-39
2	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	56,070	1360	960	7-3/4	7	6-11/16	6-3/32	FF-1052 Class (B&W Boilers)
2a	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	53,515	1340	950	7-3/4	7	6-11/16	6-3/32	FF-1040 Class FF-1098 FFG-1-6
3	Class 2500	2-1/2	Class 300	4	Class 300	1/2	180	86,000	1375	975	7-3/4	7	6-11/16	6-3/32	CV-60-62
3a	Class 2500	2-1/2	Class 300	4	Class 300	1/2	180	81,300	1305	970	7-3/4	7	6-11/16	6-3/32	CV-63, 64
4	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	55,000	1325	950	7-3/4	7	6-11/16	6-3/32	DDG-4-6, 9, 14 DDG-15-19 DDG-23, 24
5	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	45,000	1320	950	7-3/4	7	6-11/16	6-3/32	
5a	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	45,100	1325	950	7-3/4	7	6-11/16	6-3/32	DD-946
6	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	61,750	1340	960	7-3/4	7	6-11/16	6-3/32	DDG-2, 3, 7, 8 10-13 DDG-20-22
6a	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	60,600	1340	960	7-3/4	7	6-11/16	6-3/32	
7	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	69,300	1375	960	7-3/4	7	6-11/16	6-3/32	CG-16-18 DDG-40-46

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TABLE II. *Dimensional and design data – Continued.*

Style	Inlet flange type (ANSI B16.5)	Inlet flange size (inch)	Outlet flange type (ANSI B16.5)	Outlet flange size (inch)	Drain flange type (ANSI B16.5)	Drain flange size (inch)	Drain flange location w/regard to outlet (degrees)	Flow rate (lb/hr)	Pop pressure (lb/in^2)	Steam temperature (°F)	Center to inlet (inch, plus or minus 1/16)	Center to outlet (inch, plus or minus 1/16)	Center to drain (inch, plus or minus 1/8)	Inlet to drain (inch, plus or minus 1/8)	Ship hull number
8	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	66,800	1375	950	7-3/4	7	6-11/16	6-3/32	CG-26-28, 32, 34
9	Class 1500	2-1/2	Class 300	4	Class 300	1/2	180	56,070	1360	950	7-3/4	7	6-11/16	6-3/32	FF-1052 Class (CE Boilers)
10	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	50,500	1325	950	7-3/4	7	6-11/16	6-3/32	
11	Class 2500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	89,200	1410	980	7-3/4	7	6-11/16	6-3/32	CV-66
12	Class 1500	2-1/2	Class 300	4	Class 300	1/2	90 CCW	86,500	1400	960	7-3/4	7	6-11/16	6-3/32	CV-67
13	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	92,200	710	850	6	6	6-3/4	4-19/32	CV-59
14	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	51,920	635	850	6	6	6-3/4	4-19/32	
14a	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	51,900	635	850	6	6	6-3/4	4-19/32	
15	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	180	71,300	625	850	6	6	6-3/4	4-19/32	
16	Class 600	3	Class 300	4-1/2	Class 300	1/2	90 CCW	78,250	635	850	6-13/16	7-1/4	6-3/4	5-3/32	CV-41, 43 AVT-16
17	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	39,500	685	865	6-1/2	6-15/16	6-3/4	5-3/32	AS-33, 34 AD-37, 38 LPH-10, 11
17a	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	40,594	705	865	6-1/2	6-15/16	6-3/4	5-3/32	LKA-113-116

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TABLE II. *Dimensional and design data – Continued.*

Style	Inlet flange type (ANSI B16.5)	Inlet flange size (inch)	Outlet flange type (ANSI B16.5)	Outlet flange size (inch)	Drain flange type (ANSI B16.5)	Drain flange size (inch)	Drain flange location w/regard to outlet (degrees)	Flow rate (lb/hr)	Pop pressure (lb/in^2)	Steam temperature (°F)	Center to inlet (inch, plus or minus 1/16)	Center to outlet (inch, plus or minus 1/16)	Center to drain (inch, plus or minus 1/8)	Inlet to drain (inch, plus or minus 1/8)	Ship hull number
17b	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	40,800	690	860	6-1/2	6-15/16	6-3/4	5-3/32	LPD-4-6[1]
17c	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	41,900	700	840	6-1/2	6-15/16	6-3/4	5-3/32	LPD-7-13
18	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	31,300	705	855	6	6	6-3/4	4-19/32	AE-27-29
18a	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90-CCW	29,270	700	855	6	6	6-3/4	4-19/32	AE-32-35
19	Class 900	1-1/2	Class 150	3	Class 150	1/2	180	22,509	680	865	7	5-3/4	6-3/4	5-3/32	AFS-1-7
20	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	66,100	725	865	6	6	6-3/4	4-19/32	AOE-1-3
21	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	67,500	700	860	6	6	6-3/4	4-19/32	AOE-4
22	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	48,950	700	860	6	6	6-3/4	4-19/32	AOE-5-7
23	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	39,500	705	870	6	6	6-3/4	4-19/32	AS-36, 37
23a	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	40,594	705	863	6	6	6-3/4	4-19/32	LPH-12
23b	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	41,300	710	850	6	6	6-3/4	4-19/32	LSD-36-40
23c	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	41,300	710	860	6	6	6-3/4	4-19/32	
23d	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	40,594	705	865	6	6	6-3/4	4-19/32	LKA-117
23e	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	41,900	700	840	6	6	6-3/4	4-19/32	LPD-14-15

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TABLE II. *Dimensional and design data – Continued.*

Style	Inlet flange type (ANSI B16.5)	Inlet flange size (inch)	Outlet flange type (ANSI B16.5)	Outlet flange size (inch)	Drain flange type (ANSI B16.5)	Drain flange size (inch)	Drain flange location w/regard to outlet (degrees)	Flow rate (lb/hr)	Pop pressure (lb/in^2)	Steam temperature (°F)	Center to inlet (inch, plus or minus 1/16)	Center to outlet (inch, plus or minus 1/16)	Center to drain (inch, plus or minus 1/8)	Inlet to drain (inch, plus or minus 1/8)	Ship hull number
24	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	180	40,125	705	855	6	6	6-3/4	4-19/32	LCC-19, 20
25	Class 900	3	Class 300	6	Class 300	1/2	90 Clockwise (CW)	101,283	750	915	7	8	6-3/4	5-3/32	LHA-1-5
26	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	39,500	685	865	6-1/2	6-15/16	6-3/4	5-3/32	LPH-2, 3, 7
27	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CCW	40,800	690	860	6-1/2	6-15/16	6-3/4	5-3/32	LPD-4-6[1]
28	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	41,300	700	865	6	6	6-3/4	4-19/32	LPH-9
29	Class 600	2-1/2	Class 300	4	Class 300	1/2	180	71,300	625	850	6	6		5-3/32	BB-61 Class
30	Class 600	2-1/2	Class 300	3-1/2	Class 300	1/2	90 CW	27,400	745	875	6	6	6-1/2	3-11/32	AO-177 Class

[1]LPD 4-6 – Each ship has one style 17b on one boiler and one style 27 on the other.

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3.3.1.3 **Pressure.** Unless otherwise specified (see 6.2), safety valves shall be designed and constructed to one of the standard pressures specified in 3.3.1.3.1 and 3.3.1.3.2, as applicable.

3.3.1.3.1 **600 pounds (nominal) standard.** Valves for use on boilers having drum pressures of 326 lb/in^2 to 675 lb/in^2 (for which the safety valve popping pressure may exceed 675 lb/in^2, but will not exceed 735 lb/in^2) shall be designed for a pressure of at least 735 lb/in^2 and an operating temperature of at least 875 degrees Fahrenheit (°F).

3.3.1.3.2 **1500 pounds (nominal) standard.** Valves for use on boilers having drum pressures of 676 to 1600 lb/in^2 for which the safety valve popping pressure may exceed 1600 lb/in^2, but will not exceed 1700 lb/in^2 shall be constructed for a pressure of at least 1700 lb/in^2 and an operating temperature of at least 1000 °F.

3.3.1.4 **Shut down and start up.** If maintenance requiring disassembly is required, means shall be provided to relieve all pressure-containing pockets without creating a hazard to personnel. If lay up or maintenance on the valve or its boiler can adversely affect its performance by loss of primary fluid, or other causes, the design and construction shall incorporate means for properly placing the system into service, when necessary. Venting or other means used shall not constitute a hazard to personnel. Such means used shall be foolproof to eliminate the possibility of adversely affecting normal service. Items such as petcocks shall be unacceptable.

3.3.2 **Shipboard environmental conditions.** Safety valves shall be designed and constructed to perform their required functions in a reliable manner when installed and operated on board Navy ships under the following shipboard arrangements and environmental conditions, which shall be understood to prevail simultaneously and to occur in any combination.

3.3.2.1 **Installation orientation.** The valve shall be configured for installation in either the athwartships or the fore-and-aft orientation.

3.3.2.2 **Atmosphere.** The valve shall be constantly exposed to a marine environment that shall be understood to include extremes of ambient temperatures, high humidity in an at-sea atmosphere, and conditions conducive to electrolytic corrosion of dissimilar metals.

3.3.2.3 **High impact shock.** The total safety valve system shall be designed and constructed to withstand high impact shock that might be caused by popping, firing of the ship's own armament and noncontact underwater explosions of near-miss aerial bombs, torpedoes, and mines. Under such conditions, it shall remain functionally reliable. As a result of such shocks, the valve shall cause no adverse effect to the boiler. The valve shall pass type A shock test for grade A, class I equipment in accordance with MIL-S-901. NAVSHIPS 250-660-30 may be used as a guide in the design of shock resistant Naval equipment.

3.3.2.4 **Vibration.** The valve shall be exposed to ship environmental vibrations. It shall meet the type I vibration requirements in accordance with MIL-STD-167-1.

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3.3.2.5 **Inclination.** The valve shall be exposed to cyclic and, possibly, permanent ship inclinations of a ship in a seaway and shall perform reliably under the following inclination and motion requirements:

Permanent trim by bow or stern	5 degrees
Permanent list	15 degrees
Symmetrical rolling	45 degrees
Symmetrical pitching	10 degrees

These are displacements from the horizontal position. Trim and list as well as roll and pitch conditions shall be considered as occurring simultaneously.

3.3.2.6 **At-sea performance.** Two valves shall be tested as specified in 4.6.8.2 to determine maintenance requirements and that performance, reliability and maintainability.

3.3.3 **Safety criteria.** The design and construction shall be such that all procedures for operating, inspecting, maintaining, and adjusting the valve can be effected in a safe manner. To ensure maximum protection for operating and maintenance personnel against hazards and accidents, the valve shall, as a minimum, incorporate the following safeguards and protective features:

a. Exposed moving parts that could be a hazard to personnel shall be suitably covered by removable, expanded metal guards or similar sturdy material. Such guards shall not interfere with valve operation or maintenance.

b. Exposed edges shall be generously rounded and smooth to prevent cutting edges and sharp corners.

c. Safety valve components of heavy or unwieldy nature shall be mounted so that they shall not fall, topple, slide, or drop when their mounting fasteners are removed. Lifting the components for mounting and dismounting or assembly and disassembly shall constitute a separate action apart from removing the mounting fasteners.

3.3.4 **Ease of maintenance.** Ease of maintenance shall be a consideration. Proper regard shall be given to the fact that maintenance often may be performed under adverse conditions, with the ship at sea far from normal sources of supply and may be performed by maintenance personnel who may not be seasoned mechanics. Criteria for ease of maintenance shall include parts standardization, parts interchangeability, configuration control, accessibility for maintenance and repair, and portability.

3.3.4.1 **Parts standardization.** The design and construction shall standardize those components, fasteners, and items used in order to reduce the number of sizes and kinds of items that are generally similar. For example, in a design incorporating the same size bolts in two locations shall standardize on the same bolts, even if it shall mean upgrading the bolts at one of the two locations. Wherever practicable, the contractor shall design and construct for the use of standard, commercially available parts. Seals and gaskets shall be standardized to the maximum extent possible and shall be of a configuration and type commonly available from industrial supply houses.

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3.3.4.2 Interchangeability. Identical components and parts, to include repair parts, furnished on the same contract or order or built to the same drawings shall be interchangeable as manufactured. Components which are functionally interchangeable shall, as far as practicable, be physically and mechanically interchangeable, as applicable. The unloader valve and pilot valve assembly shall be constructed for modular replacement of parts by ship's personnel. However, when two similar items are not interchangeable, functionally and strengthwise, they shall not be physically interchangeable and shall differ significantly so that their non-interchangeability shall be apparent and shall assure correct installation.

3.3.4.3 Configuration control. Components and their parts shall be constructed so that replacement parts subsequently constructed in accordance with the contractor's drawings shall be replaced without additional machining, handfitting or otherwise customizing (see 6.3 and Appendix A).

3.3.4.4 Accessibility for maintenance and repair. These valves shall be constructed so that degree of disassembly for maintenance and repair shall be held to a minimum. Wherever possible, disassembly and reassembly methods shall require the use of standard tools. Cutting, welding, brazing, and allied processes shall not be required for disassembly and reassembly associated with maintenance.

3.3.4.5 Portability. Proper construction consideration shall be given to the maximum weight, bulk, size, and shape that maintenance personnel can lift and transport safely during repair on board ship. Components that are not easily transported because of weight, shape, or finish shall be provided with padeyes, lifting lugs, or the equivalent.

3.3.5 Assembly processes.

3.3.5.1 Cleanliness. Components, castings, weldments and machined parts, particularly their internal passages, shall be thoroughly cleaned and inspected for cleanliness prior to assembly. After assembly, internal passages shall be purged clean of all construction debris. During assembly and upon completion, openings where contaminants could enter shall be sealed or otherwise protected appropriately.

3.3.5.2 Alignment. Components and assemblies that are alignment-critical shall be positively held in place by dowel pins, fitted bolts, keys, rabbeted or tongue and groove joints, or other indexing means, as appropriate. The construction shall ensure that the alignment shall not be disturbed or undue stresses be induced in service as a result of the alignment means.

3.3.5.3 Threaded parts. Threaded parts shall conform to FED-STD-H28 and FED-STD-H28/1, FED-STD-H28/2, FED-STD-H28/4, FED-STD-H28/6, FED-STD-H28/19, and FED-STD-H28/21. Screw threads used in design and construction of these valve assemblies shall be in accordance with the Unified National Coarse (UNC) series class 2 or 3 fit, except for studs that shall be a class 3 fit, in accordance with FED-STD-H28. Through bolting shall be the preferred method of joining threaded parts whenever possible. If not possible, then the use of studs shall be preferred to the use of capscrews or tapbolts. Bolt-studs and studs shall be sufficiently long to insure that, when nuts are tightened, there shall be at least one thread projection above the outer face of the nut. Tapered pipe threads shall not be used as connections for piping or to components (see 6.2).

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3.3.5.4 **Welding.** Welding and allied processes, to include requirements for radiographic and magnetic particle inspection, shall be in accordance with MIL-STD-278.

3.4 **Component construction.** The provisions of this section cover requirements for the design and construction of the components of the boiler safety valve including materials specific to any components. The provisions herein shall not relieve the contractor from meeting other requirements of this specification.

3.4.1 **Unloader valve.**

3.4.1.1 **Dimensions.** Unloader valves shall meet the dimensional and design requirements for safety valves specified in NAVSHIPS 5000-S5103-841622 and in table II, for the applicable style and class. Valves shall be designed and constructed for modular replacement.

3.4.1.2 **Valve nozzles (seat bushings) and discs.** Valve nozzles shall be of the "full nozzle" or "through bushing" type for all operating pressures. If the valve disc is guided from beneath, the guide wings shall extend below the seat to a distance not less than one half of the valve diameter. Guides shall be supported in such a way as to ensure consistent operation and tightness and to ensure that steam force, while the valve is blowing, shall not cause binding or seizing of the valve. The unloader valve shall be a pressure seating design and construction in which the drum boiler water pressure acting on a piston keeps the valve tightly seated until valve lifting is initiated automatically by the pilot valve or by the manual means provided. During hydrostatic testing, water temperature shall not exceed 100 °F. The valve shall be constructed to attain full lift at a pressure not higher than 3 percent above the set pressure.

3.4.1.3 **Valve gag.** The valve shall contain a built-in gag that prevents unloader valve operation. A means shall be provided to permit a visible determination as to whether the valve is in the gagged or non-gagged position. A locking mechanism shall be provided to prevent gagging the valve without first opening the lock.

3.4.1.4 **Radiographic and magnetic particle requirements.** Valve body parts that may be subjected to full boiler steam pressure shall meet the radiographic inspection requirements of MIL-STD-278.

3.4.2 **Remote pilot assembly.** The remote pilot assembly shall consist of those components remotely located from the unloader valve and shall include the remote pilot valve, the canister, the filter, the manually operated three-way valve, associated piping and necessary appurtenances. Valve settings shall be controlled from the pilot assembly and changes in settings shall be made without the need to touch the unloader valve. The pilot assembly shall be made as compact as is possible and shall be a separate, self-contained unit. The pilot assembly shall be located in and fixed to a rigid mounting plate in a metal cabinet that shall contain means for attachment to a bulkhead. The cabinet interior shall be ample to permit access for adjustments and maintenance. A clear, well-prepared drawing of how the safety valve and remote pilot assembly shall be connected to the boiler shall be photoetched on a plate at least 12 by 8 inches and installed inside on the door of the pilot assembly cabinet. The plate shall also contain the valves required in paragraph 3.6 d,e,f,g, and h. The cabinet shall be painted with two coats of Navy gray paint and shall contain means to permit locking.

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3.4.2.1 Remote pilot. The remote pilot shall be a multi-function valve and control system which shall:

a. Function using the boiler fluid as the hydraulic fluid

b. Continuously sense the pressure in the boiler steam drum and act to maintain pressure on the unloader valve piston to keep it seated tightly during normal conditions

c. Automatically cause the unloader valve to open fully when the set pop pressure is reached or exceeded

d. Automatically cause the unloader valve to reseat at the set reseating pressure after blowdown

e. Cause the unloader valve to fully open, blowdown and reseat when actuated from a hand-operated lifting gear

f. Incorporate other means to perform as specified herein.

3.4.2.1.1 Hand-easing gear. The pilot shall be provided with a lever for the attachment of a hand-easing gear so that the three-way valve may be operated to actuate fully opening the unloader valve, as required, and to allow the unloader valve to close automatically upon release of the force on the easing gear. The three-way valve shall also return automatically to its normal operating position. In order to prevent damage to any part of the valve, the pilot valve shall have incorporated in this lifting mechanism a device to prevent lifting the valve by hand beyond the maximum lift of the valve. Unless otherwise specified in the contract or order (see 6.2), the valve shall be fully opened by 1-1/2 inch movement of the lifting mechanism with not less than 150 or more than 180 pounds of lifting force on the lever.

3.4.2.1.2 Alternate gag. Provisions shall be made in the pilot valve to allow installation of a temporary clamping device that, when installed, shall prevent the safety valve from lifting, thus acting as an alternative means of gagging the valve. Visible means shall be provided for showing clearly whether the valve is in the gagged or non-gagged position.

3.4.2.1.3 Locking mechanism. Locking mechanisms shall be provided that shall prevent changes in valve settings and prevent gagging the valve shut without first opening the locks.

3.4.2.1.4 Drain connections. Flanged end drain piping connections of 1/2-inch pipe size shall be provided in accordance with NAVSHIPS 5000-S5103-841622 and table II, herein. The drain connection orientation shall be as specified in the contract or order (see 6.2).

3.4.2.2 Piping and associated equipment. Piping for control systems shall be hard stainless steel tubing, grade 304 or 316, in accordance with ASTM A 269. Mechanical tubing connections shall be machine thread union type with O-ring seal. Piping, flanges and fittings that may convey steam shall be in accordance with MIL-STD-777. Spiral wound gaskets may be of the inner and outer ring construction. Tapered pipe thread connections shall not be permitted.

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3.4.2.3 **Manual control valve.** The manually operated three-way valve shall be suitable for operation with boiler water up to 3000 lb/in^2 and a temperature up to 1000 °F. Unless another type of three-way control valve is determined to be acceptable as a result of first article testing and examination, the three-way valve shall be Teledyne Republic Valve number 8093E-WS28 or equal. Other types of three-way control valves proposed shall be physically interchangeable with Teledyne Republic Valve number 8093E-WS28.

3.4.2.4 **Springs.** The permanent set of the springs, 10 minutes after being released from a cold compression test which closes the spring solid, shall not exceed 0.015 inch. The springs shall not be compressed solid when the valve is at its maximum lift.

3.4.2.5 **Spring range.** Springs shall be constructed so that the popping pressure can be varied over a range of approximately 10 percent of the ranges specified in 3.5.2.1 and 3.5.2.2 respectively.

3.4.2.6 **Canister and filter.** The canister and filter assembly shall remove particulate matter that could interrupt pilot valve operation. Flow capacity through the canister and filter shall be sufficient (even with a 20 percent reduction in filtration area caused by clogging) to permit blowdown and reseating as specified herein.

3.5 **Performance.**

3.5.1 **Relieving capacity.** The relieving capacity of the pilot operated safety valve shall be in accordance with the ASME Boiler and Pressure Vessel Code, section I or as specified in the contract or order (see 6.2), and shall be based upon 103 percent of popping pressure. The tolerance allowed for the actual capacity of each valve shall be minus zero and plus 5000 pounds per hour (see 6.3).

3.5.2 **Safety valve settings.**

3.5.2.1 **Popping pressure.** The pilot actuator valve shall be designed and constructed to pop and open the superheater safety valve at a pressure of 8 percent greater than the highest working steam drum working pressure (120 percent overload). The low drum safety valve shall be set to pop at 3.5 percent of the highest working steam drum pressure higher than the pilot valve actuator valve setting. The valve shall be designed and constructed so that the popping pressure is adjustable and able to be set at any pressure within a range from 5 below to 5 percent over the popping pressure. Popping shall be achievable within plus or minus 5 lb/in^2 of the set popping pressure. Valve opening shall be immediate, positive, and characterized by a full opening pop action with no simmering. Prior to delivery, valves shall be set at the popping pressure as specified in the contract or order (see 6.2) and verified by popping.

3.5.2.2 **Blowdown.** After opening, the valve shall remain fully open during blowdown until the pressure has dropped to the specified, preset reseating pressure; at that point, the valve shall close sharply and tightly without chattering or simmering. Unless otherwise specified, the reseating pressure shall be within the range of 3 to 4 percent below the popping pressure. Prior to delivery, valves shall be set for the reseating pressure specified in the contract or order (see 6.2) and

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verified by blowdown. Unless other allowable tolerances are specified in the contract or order (see 6.2), reseating shall be achievable within plus or minus 5 lb/in^2 of the set reseating pressure. Blowdown shall be within the allowable limits determined by the specified setting range of the valve. Valves shall be designed and constructed for the specified range of blowdown.

3.6 Marking. Safety valves shall be plainly marked by the contractor. The markings shall be stamped or cast on the body or stamped on an identification plate permanently secured thereto that shall be in accordance with MIL-P-15024 and MIL-P-15024/5. Identification plates shall contain the following information:

a. Name or identifying trademark of contractor and designating number or symbol
b. The contractor's model or type number and serial number, if applicable, that can positively identify the item
c. The nominal size, and nominal operating pressure for which the valve body is suitable (preferably cast on the valve body, for example, 2-1/2 - 600)
d. The authorized popping pressure
e. The authorized reseating pressure
f. The rated relieving capacity, as specified in 3.5.1
g. Vertical lift of valve necessary to give rated relieving capacity
h. MIL-V-24740, class and style (see 1.2 and 1.2.1).

3.7 Special tools. The need for using special tools for adjustments or maintenance shall be minimized. Common tools and wrenches listed in the Federal Supply Catalog shall not be furnished. A copy of this catalog is available for review in the local office of the Defense Contract Administration Services Management Area (DCASMA).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

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Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the examination and tests as specified in 4.5 and 4.6. Three safety valves of each class and style shall undergo first article inspection as specified in table III (see 6.3).

TABLE III. *First article quality conformance inspection.*

Examination and tests	First article	Quality conformance
Examination	4.5	-
Hydrostatic	4.6.1	4.6.1
Steam tightness	4.6.2	4.6.3
Shock	4.6.4	-
Vibration	4.6.5	-
Popping and reseating pressures	-	4.6.6
Radiographic and magnetic particle inspection	4.6.7	4.6.7
Performance	4.6.8 through 4.6.8.2.1	-

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table III (see 6.3).

4.4.1 **Sampling for quality conformance.** Boiler safety valves (including components) of the same size, class and style presented for delivery at one time shall be considered a lot for purposes of quality conformance inspection. Samples shall be selected from each lot.

4.4.2 **Action in event of failure.** If a valve fails a requirement, the valve shall be segregated and the reasons for failure recorded and documented. The proposed remedial actions shall be agreed on by the acquiring activity prior to re-examination or re-testing. The contractor shall identify whether corrective action would affect performance or other characteristics previously approved as a result of first article testing.

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4.4.2.1 Re-examination. If a sample fails the examination, each valve in the lot shall be examined and shall have the deficiencies satisfactorily corrected and shall pass the examinations of 4.5 to determine acceptability. Defects shall be as specified in table IV.

TABLE IV. *Classification of defects.*

Categories	Defects
Critical:	None defined
Major	
101	Valve not style, class, or size specified
102	Valve not complete; component parts missing
103	Component parts not new; evidence of reclaimed, used or salvaged components
104	Evidence of use of unauthorized materials or repairs
105	Threaded parts nonconforming; not specified class, fit; evidence of burring, crossing, stripping
106	Threaded projection from outer face less than required
107	Springs not of noncorrosive material
108	No stop provided to fix position of valve after adjustment
109	Valve nozzles not welded into valve body; not "full nozzle" or "through bushing" type, as specified
110	Lock mechanism does not prevent altering spring adjustment without opening lock
111	Hand-lifting gear not provided
112	Drain connection size and location nonconforming
113	Similar parts not interchangeable
Minor	
201	Markings nonconforming; not stamped or cast on body, or stamped on identification plate permanently secured to body; not permanent; not legible; data incomplete
202	Specified repair parts not furnished
203	Specified special tools and accessories not furnished
204	Specified drawings not furnished; not specified type; incomplete, incorrect
205	Specified technical manuals not furnished; incorrect, incomplete, inadequate

4.4.2.2 Re-testing. If a sample had previously failed to pass a test, each valve in the lot shall have the deficiencies corrected and shall pass the tests in 4.6.

4.5 Examination. After completion of first article tests, the valves shall be examined for signs of weakness, incipient cracks or other disruptions which might harm personnel or create other hazards. The visual examination shall include looking at dimensions and materials, checking for cleanliness of parts and missing parts, and verifying correct assembly and alignment.

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4.6 **Test procedures.** Shock testing (4.6.4) and vibration testing (4.6.5) shall be conducted prior to the other tests and examinations required.

4.6.1 **Hydrostatic test.** The valves shall be held closed by placing a gag on the valves hand tight (see 3.4.1.3) and then shall be subjected to a water pressure of 1-1/2 times the popping pressure of the highest set pressure of the valves. There shall be no leakage between the nozzle (seat bushing) and disc or through the body of the valve.

4.6.2 **Steam tightness (first article).** The steam boiler used to supply steam for testing shall be of sufficient capacity to cover the full pressure drop necessary to achieve blowdown requirements for the valve concerned. The procedure outlined below shall be considered the standard for determining steam tightness. Valves tested shall be free from preservative prior to testing. After a suitable warming-up period on the test stand, the valve shall be popped, reseated, and adjusted to the proper popping pressure and blowdown within the tolerances specified in 3.5.2.1 and 3.5.2.2. The valve shall then be popped by steam fifty times, after which a steam-tight cover shall be affixed to the exhaust flange and the line connected to a cooling condenser. The condensate from the condenser shall be collected for measurement. Measurement shall be conducted as follows:

a. Steam pressure shall be raised to 90 percent of set pressure.

b. The condensate shall be measured from the above system for a period of 30 minutes to determine if steam tightness requirements have been met. Any leakage shall be measured from the pilot panel. There shall be no visible steam leakage between the valve spindle and its guide. The measured leakage rate (total of the condensed leakage from the valve and any boiler water leakage from the pilot assembly) shall be not greater than that specified.

4.6.3 **Steam tightness (quality conformance).** The steam source used to supply steam for testing shall be of sufficient capacity to cover the full pressure drop necessary for the valve concerned. The procedure outlined below shall be considered the standard for determining steam tightness for delivery. Valves tested shall be free from preservative prior to testing. After a warming-up period on the test stand, the valve shall be set at the required settings within the tolerance specified in 3.5.2.1 and popped by steam five times, after which a steam-tight cover shall be affixed to the exhaust flange and the line connected to a cooling condenser. The condensate from the condenser shall be collected for measurement. Measurement shall be conducted as follows:

a. Steam pressure shall be raised to 90 percent of set pressure.

b. The condensate shall be measured from the above system for a period of 15 minutes to determine if steam tightness requirements have been met. Any leakage from the pilot valve assembly panel drains shall be measured.

The leakage rates above shall be the total of the condensed leakage from the valve and boiler water leakage from the pilot valve assembly. Steam tightness acceptance shall be based on the same criteria as in 4.6.2.

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4.6.4 Shock. High impact shock tests for the unloader valve and the pilot assembly shall be conducted in accordance with type A of MIL-S-901 for grade A, class I equipment. The valve shall be considered to have met the high impact shock requirements if the following conditions have been met:

a. No breakage of any part, including accessories and bolting shall have occurred. Examination shall reveal no incipient cracks or signs of weakness that could result in failure if operated.

b. The valve shall have performed and shall continue to perform without maintenance or adjustment.

c. There shall have been no dislocation of components or appurtenances which could render the valve unsafe for continued service.

4.6.5 Vibration. Type I vibration tests for pilot valve assembly and superheater (unloader) valve shall be conducted in accordance with MIL-STD-167-1. The valve shall be considered to have met the vibration test requirements if it continues to perform accurately during and after the vibration tests and shows no evidence of weakness or disarrangement.

4.6.6 Popping and reseating pressures. Each valve shall be popped at three popping pressures to cover the setting range of the spring (see 3.5.2.1). The settings shall be chosen to assure that the range of setting shall be obtained. No simmering shall be permitted at any set pressure. Valve closure shall be clean and sharp. Failure shall be defined as follows:

a. Full popping range not covered
b. Lifting and reseating pressure nonconforming
c. Time lag between lifting and reseating on pilot and low drum valves nonconforming
d. Popping and reseating are not clean and sharp.

After completion of these tests and prior to packaging, each valve shall be adjusted to the settings required by the contract or order (see 3.5.2) and verified by test.

4.6.7 Radiographic and magnetic particle inspection. Portions of the valve that may be subjected to full boiler pressure while in service shall be subjected to radiographic and magnetic particle inspection in accordance with MIL-STD-278. There shall be no evidence of pockets, porosity, occlusions, slag, hard spots, or discontinuities.

4.6.8 Performance. The valves shall be set to pop and reseat at the popping and reseating pressures specified in the contract or order (see 6.2), when the boiler is under static load (no steam flow through the main steam outlet). Superheater safety valves shall be set with steam having superheat not in excess of 50 °F. After such setting, the boiler shall be brought up to the rating necessary to hold the valve open and to obtain the specified steam temperature at the superheater outlet. This rating shall be held for not more than 1 hour before popping each valve. To evaluate temperature effect on performance, each valve shall then be popped with the boiler under running load (steam flowing through the main steam outlet), held open for 2 minutes and then reseated. The popping and reseating pressures under this condition shall not vary from the

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settings specified in the contract or order by more than plus or minus 1 percent of the reseating pressure. Immediately after reseating, the valve shall be popped again and allowed to reseat. The time interval before popping again shall be only that required to control the boiler pressure. The variations in popping pressure under this condition from the specified popping pressure shall not exceed 1-1/2 percent of the specified reseating pressure, and the variation of the reseating pressure from the specified reseating pressure shall not exceed 1 percent of that pressure. Reduction in blowdown within the above tolerances shall not affect the proper functioning of the valve.

4.6.8.1 **Sequence time.** In addition to the tests specified in 4.6.8 (or in conjunction therewith), runs shall be made to ascertain the time lag between popping and reseating of the pilot section and the lifting and reseating of the unloader valve. Under static load, with the main stop secured, the interval between opening of the pilot valve and lifting of the unloader valve shall not exceed 2 seconds. Similarly, the interval between reseating of the pilot valve and reseating of the unloader valve shall not exceed 2 seconds. Under running load with specified steam temperature at the superheater outlet, the time lag shall not exceed 2 seconds in either case.

4.6.8.2 **At-sea performance test.** After the valve has passed all other first article tests, arrangements shall be made through NAVSEA to install a minimum of two valves on a test ship. The valves shall be set to required pressure and popped by steam to confirm the setting. The valves shall be left on the ship for a period of at least 9 months to experience normal operation. The valves shall then be inspected and evaluated for performance history, maintenance requirements, and reliability by the Naval Ship Systems Engineering Station, Philadelphia, PA. Any performance, maintenance, reliability, or material problems shall be corrected by the contractor, and the valves shall experience an additional period of actual trouble-free service for a period of 6 months prior to acceptance of the valves as fulfilling the requirements of this specification.

4.6.8.2.1 **At-sea performance test requirements.** The purpose of the at-sea performance test is to confirm that the new system maintains its availability to function properly over a prolonged period of time in the shipboard environment. Valve performance shall be considered acceptable if:

a. Valve settings are maintained within plus or minus 5 lb/in^2 of original required settings.

b. Valve functions properly; lifts and reseats without chatter or simmer.

c. When retested in the laboratory after shipboard service, valve displays no visible leakage on ship and meets the requirements of 4.6.2.

d. No malfunctions are experienced during the testing period or all malfunctions have been corrected and the system operated satisfactorily as indicated above for a 4-month period or longer as prescribed by NAVSEA.

e. Replacement of parts caused by wear or imminent malfunction shall not be required after the shipboard test. Allowances will be made for normal wear.

4.7 **Inspection of packaging.** Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packing requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, (see 6.8.)

5.1 Packaging requirements. Safety valves and tools shall be preserved level A, C, or commercial, packed level A, B, C, or commercial as specified (see 6.2) and marked in accordance with MIL-V-3 and MIL-S-196 respectively and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

a. **Navy fire-retardant requirements.**

(1) *Lumber and plywood.* Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	–	Type II – weather resistant Category 1 – general use
Level C	–	Type I – non-weather resistant Category 1 – general use

(2) *Fiberboard.* Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The boiler safety valves covered by this specification are intended for installation at the superheater outlet or on the steam drum with the pilot assembly located remotely from the boiler and unloader valve. This provides features for ease of maintenance, shipboard level repair capability for satisfactory replacement of moving parts, high service time to leakage rate ratio, simplified ease of setting features, cold pilot setting ability, and a built-in gagging feature. The valve uses boiler fluid as the working fluid and operates completely without any auxiliary means of power such as air or electricity.

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6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Class and style of valve (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
d. When first article inspection is required (see 3.1)
e. Design pressure, if other than specified (see 3.3.1.3); (Note that the pressure class of the inlet flange, as shown in table II, does not necessarily coincide with the valve design pressure)
f. Material of bolt-studs and studs, if other than specified (see 3.3.5.3)
g. Lifting mechanism movement and force, if other than specified (see 3.4.2.1.1)
h. Drain connection orientation (see 3.4.2.1.4)
i. Relieving capacity, if other than specified (see 3.5.1)
j. The popping pressure setting, if other than specified (see 3.5.2.1)
k. The reseating pressure setting and tolerances, if other than specified (see 3.5.2.2)
l. Popping and reseating pressures (see 4.6.8)
m. When fire-retardant treated materials are not required (see 5.1)
n. Level of preservation and packing required (see 5.1)
o. Special marking required (see 5.1).

6.3 **Consideration of data requirements.** The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.2.3	DI-E-2121	Certification of compliance	–
3.3.4.3 and Appendix A	DI-DRPR-80650	Engineering drawings	–
3.5.1	UDI-E-23213	Diagrams, calculations and stress	–
4.3	DI-T-4902	First article inspection report	–
4.4	DI-T-5329	Inspection and test reports	–

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Acceptable quality levels. Quality levels considered acceptable for boiler safety valves (including components) are those for samples meeting MIL-STD-105, inspection level III, acceptable quality level (AQL) 0.65.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.7 Subject term (key word) listing.

At-sea performance
Blowdown
Inclination
Popping pressure
Relieving capacity
Shut down and start up

Preparing activity:
Navy - SH
(Project 4820-N026)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80650 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. When required by the contract or order, drawings shall contain the following information:

30.1.1 General requirements.

30.1.1.1 "Manufacturer's use only" notes. Information intended for manufacturer's use only shall be so designated. On a commercial drawing with wide usage, it is permissible to designate portions thereon "For Navy Use Only."

30.1.1.2 Dimensional tolerances. Unless otherwise indicated by note, legend, or key, all tolerances shown on a drawing shall be complied with during both manufacture and maintenance repair.

30.1.1.3 Identification of purchased items. Prime contractors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the item(s) delineated thereon are produced by the subcontractor. The prime contractor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

30.1.1.3.1 Semi-finished parts. Prime contractors who purchase semi-finished parts from subcontractors for final production, test or selection phases in their own shops have the option of using, as the single drawing identification, either their own title block and drawing number or the title block and drawing number of the subcontractor but not both.

30.1.1.4 Associated equipment. Drawings of associated equipment shall be in accordance with the specifications covering those equipments.

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30.1.1.5 **Special processes.** Where items require a special or unique process in manufacture or assembly, the appropriate drawings shall identify the nature of the process. Drawings showing the application of studs requiring the use of a locking resin shall have a notation requiring: (a) manufacturer's recommendation regarding the use of primer; (b) manufacturer's recommended primer; and (c) the minimum and maximum waiting times associated with the use of the primer and resin. The notation shall require strict adherence to these requirements.

30.1.2 **Design drawings.** Design drawings shall consist of the drawings described in 30.1.2.1 through 30.1.2.4. Basic design drawings consist of the drawings described in 30.1.2.5 and 30.1.2.5.1.

30.1.2.1 **Sectional assembly drawings.** Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts. All running clearances shall be labeled and dimensioned as diametral clearances (the difference between diameters of mating parts). This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view. Tightening torques with tolerances and thread lubrication requirements for threaded fasteners shall be shown on the assembly drawing.

30.1.2.2 **List of materials.** The list of material shall include every part required in the assembly, including those parts not required to be detailed. The list of material shall include an indication of each part required by this specification to be furnished as an onboard repair part (this is not a repair parts list). The list of material shall include the manufacturer's drawing number and service part number for all parts for which detail drawings are required (see 30.1.2.3), as well as quantity required, material type, and material specification for all parts. Parts shall be named to indicate the function they serve. Parts for which detail drawings are not required shall be identified as to dimensions. The list of material may be shown on the sectional assembly drawing.

30.1.2.3 **Detail drawings.** Detail drawings shall be prepared for all parts and sub-assemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance and overhaul. Details of these parts shall be so complete as to permit emergency manufacture by a naval shipyard without assistance from the original manufacturer. Details of any castings shall be furnished. For all cast parts, the minimum section thicknesses shall be dimensioned and toleranced. Sub-assemblies whose parts cannot be acquired or serviced individually, shall be shown as a single part and so indicated. Multi-detail drawings are preferred, but mono-detail drawings may be used.

30.1.2.4 **Outline drawings and certification data.** Outline drawings and certification data shall be in the form of supplemental drawings to all sectional assemblies and detail drawings. Separate outline drawings, drawing lists, and certification data shall be prepared unless the complete equipment covered by the outline drawing and the referenced drawings are in fact identical in all respects.

30.1.2.4.1 **Certification data sheets.** Certification data sheets shall include a table on spring data flow (through unloader valve and through pilot valve), authorized valve settings, and blowdown. In addition to the authorized pressures for which the valve is set, the adjustable ranges of popping and reseating pressures also shall be shown.

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30.1.2.4.2 Outline drawings. The outline drawings, in addition to the certification data required, shall contain the following:

a. Dimensional outline assembly drawing of the valve and its pilot assembly. Also include the weight of each main component and the total weight of the valve.

b. Endurance tests, high-impact shock tests and nondestructive tests (if performed) and the NAVSEA letters approving these reports or extension of any tests.

c. Overall dimensions of the complete unit including location of all fittings and end connections, the space required for removal and replacement of parts for maintenance, and mounting information. The drawing shall include dimensions and a tolerance for each of the following dimensions. Each dimension shall also be shown with the same value and tolerance on the detail drawing for the part. For all flanged piping connections show:

(1) The diameter of the bolt circle and the concentricity of that circle relative to the axis of the connection

(2) The angular location of each bolt hole with respect to some reference line shown on the drawing. Even if the angle to the reference line is zero or 90 degrees, that number shall be stated, with a tolerance. It is acceptable to locate only one bolt hole relative to the reference line and to state the remaining holes are equally spaced as long as a tolerance is specified on the equalness of the spacing.

(3) In lieu of (1) and (2) above, an X-Y coordinate system may be used to locate the bolt holes.

(4) The diameter of the bolt holes

(5) The tolerance on perpendicularity between the flange face and the axis of the connection

(6) The inner diameter of the suction and discharge nozzle at the flange face and the concentricity of that diameter relative to the axis of the connection.

d. Identification of any special tools required.

30.1.2.5 Basic design drawings. Basic design drawings covering valves under this specification should be submitted to NAVSEA prior to and independent of invitations for bids. As such, these are different from design drawings as described in 30.1.2. Each drawing may cover a range of sizes of valves so long as they are of the same basic design. Detail drawings of individual parts are not required in connection with basic design drawings, but may be included, if desired by the manufacturer for clarification, or to obtain advance tentative approval of NAVSEA.

30.1.2.5.1 Content of basic design drawings. Basic design drawings shall consist of small scale (not necessarily to scale) plan and elevation views showing overall dimensions. If the plans cover a family of valve sizes all of the same design, the dimensions may be tabulated. The range of capacities, range of settings, and pressures shall be shown. The main part of this drawing shall be

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an undimensioned sectional assembly with complete list of material which shall include every part required in the valve assembly. If materials of some parts change with various valve characteristics, a table shall be included to show these variations. Alternative subassembly arrangements of different parts may be included, as desired.

30.1.3 Material specification identification.

30.1.3.1 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced, the following is the order of preference:

a. Federal specification or standard
b. Military specification or standard
c. Industry and technical society specification or standard
d. Manufacturer's specification or standard.

30.1.3.2 Material substitutions. Where materials other than those covered by a., b., and c. of 30.1.3.1 are referenced and approved by NAVSEA, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military Specification, in order to enable Naval Repair Facilities to make emergency repairs as necessary from Navy material stocks.

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APPENDIX B

MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included in the manuals when required by the contract or order. This in not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Standard. The following standard forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARD

MILITARY

DOD-STD-2147 Technical Repair Standards (TRS); Hull, Mechanical, and Electrical (HM&E), Preparation of; (Metric)

(Unless otherwise indicated, copies of federal and military specifications and standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

30. MANUALS

30.1 Manuals. When required by the contract or order, manuals shall contain the following information:

30.1.1 When specified in the contract or order technical manuals shall be prepared in accordance with the data ordering documents included in the contract or order. The modifications and clarifications specified in 30.1.2 through 30.1.4 shall apply. A separate manual shall be prepared for each different valve unit except in special cases when manuals covering more than one valve may be approved. Preliminary manuals shall be included for approval and shall include all proposed sections which shall be complete. Unless otherwise specified in the contract or order, manuals shall include valve controls and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

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30.1.2 Manual content. The manual shall contain, as minimum, the following:

a. *Description of system.* Components and replacement parts shall be described.

b. *Setting and operation.* Complete instructions for setting shall be included. Principles of operation and how the valve works during an actuating cycle shall be included with diagrams.

c. *Maintenance and inspection.* How, where, and when to inspect working parts. Include instructions for replacement of consumable parts.

d. *Disassembly and reassembly.* Instructions with sketches and photographs shall be included.

e. Hydrostatic testing

f. Troubleshooting guide

g. List of special tools.

30.1.3 Technical manual illustrations. Each manual shall include not less than the following illustrations covering the valve and pilot:

a. Sectional assembly drawing

b. Outline drawing

c. Complete list of material corresponding to the sectional assembly drawing

d. Certification data

e. A minimum of two photographs of the complete unit, valve, and pilot taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied, photographs of one design unit only need be supplied.

30.1.4 Technical manual quantity and distribution. The quantity and distribution of manuals shall be as specified in the contract or order.

30.2 Technical repair standard. When required by the contract or order, technical repair standard (TRS) shall contain the following:

30.2.1 Technical repair standard content. TRS technical content shall include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment's dimensions, clearances, and tolerances such that the equipment is capable of performing its function as originally specified and is capable of being logistically supported by the Department of Defense Logistics support system.

30.2.2 Technical repair standard acceptibility. The TRS format and content shall conform to DOD-STD-2147 and shall be approved and validated in accordance with this standard.

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